

AI-Assisted UX Research for an Automated Personal Expense Tracking App

Research frame and viability

This exploration is well-suited to an **AI-assisted early-discovery approach** because the product concept sits at the overlap of three realities: people make many small payments each month, budgeting discipline often breaks down under repeated manual effort, and privacy concerns rise sharply when financial data moves outside first-party banking channels. In consumer budgeting research, 62% of surveyed consumers said they had created a budget, but only 48% maintained one regularly; the same study found that mobile is the primary budgeting platform for 73% of respondents. Separately, the Atlanta Fed’s payment diary reported an average of 47 payments per consumer per month, and recent U.S. diary findings show cards remain dominant by payment count. Taken together, that means even modest input friction compounds quickly in real life. [1]

For an Iran-first concept, the **platform implication is decisive**: this product is strongest as an **Android-first experience**. Statcounter’s May 2026 figures for Iran show Android at 82.96% mobile OS share versus 14.57% for iOS. On Android, notification-listener access can expose posted notifications after explicit user grant, but Google Play heavily restricts READ_SMS, and Android 15 now masks one-time-password content for most notification listeners. On iOS, Apple’s public notification APIs are centered on an app’s own notifications, while cross-app notification forwarding is documented for accessory companion use cases rather than general-purpose finance apps. The practical design inference is that **automatic extraction from banking notifications is much more viable on Android than iPhone**, and especially more viable than SMS-reading for a Play-distributed app. [2]

The local signal source is also **real but heterogeneous**. Bank Melli publicly documented that some low-value alerts were moved off SMS into official bank apps from March 2022, which suggests a notification-driven tracker cannot assume the same message pattern across banks, transaction sizes, or channels. In addition, the June 2026 cyberattacks affecting major Iranian banks make the security climate unusually salient for any product that asks for notification access or aggregates transaction history. That does not invalidate the opportunity; it means trust design has to be first-class, not decorative. [3]

Consumer privacy research reinforces the same point. A De Nederlandsche Bank working paper on financial-app data sharing found a real “privacy paradox” around sharing financial data with third parties, while a Government of Canada survey on open banking reported that 59% believed it would make them more vulnerable to fraud and the same share believed it would increase identity-theft risk. This aligns closely with public forum discussions in budgeting communities, where users repeatedly ask whether aggregators, connectors, or third-party finance apps are worth the privacy tradeoff. [4]

Synthetic participant dataset

I generated a **synthetic discovery dataset of 60 participants** designed to resemble raw early-stage UX research material rather than polished personas. The sample intentionally mixes students, early-career salaried workers, mid-career employees, freelancers, small business owners, traditional banking users, fintech power users, and family budget managers. The synthetic structure was calibrated against broad real-world patterns: card-heavy payment behavior, strong mobile usage for money management, and persistent privacy sensitivity around third-party financial access. It is useful for ideation and pattern-finding, but it should **not** be treated as a prevalence study or a substitute for live user validation. [5]

The full research assets are downloadable here:

- [Download the full workbook](#)
- [Synthetic participants CSV](#)
- [Synthetic participants JSON](#)
- [Real-world signals CSV](#)
- [Synthetic topline CSV](#)

The workbook includes participant profiles, survey responses, interview responses, synthetic quotes, the real-world signal-mining sheet, a theme summary, and a source index. The synthetic sample spans ages 19 to 61, with a mean age of 35.2. In the generated data, the **median participant uses 3 bank cards**, 83% use at least 2 cards, 60% use at least 3, and 80% use at least 2 banking apps. Those numbers are intentionally plausible for a fragmented-card environment and directionally consistent with the broader reality that many consumers carry and use multiple cards. For external calibration, U.S. Experian data reports an average of 3.7 credit cards regularly in use. [6]

In the synthetic survey results, 47% reviewed spending **weekly or more**, 37% reviewed it on a **monthly-ish** cadence, and 17% were clearly **reactive**, checking only after payday, when money felt tight, or after a surprise expense. Only 20% looked like active structured tracker users; 32% primarily relied on manual methods such as notes or spreadsheets, and 18% had already abandoned at least one expense-tracking workflow. On notification-driven automation, opinion was split almost evenly: 33% were ready to trust it, 35% were open but conditional, and 32% were not ready to trust it. The dominant synthetic concerns were **accuracy first**, then **privacy**, then **security**.

Synthetic distribution of bank cards used

What matters most is not the exact percentage, but the structure of the responses. The synthetic sample consistently produced a **triangle of friction**:

- money is spread across multiple cards and apps,

- manual logging fails under real routines,
- and full automation is attractive only if trust and correction are easy.

A few representative synthetic interview quotes illustrate the pattern:

- “My bank notifications make me aware, not organized.”
- “Manual tracking works until life gets busy.”
- “Reading bank notifications feels useful, but also invasive.”
- “I want the app to do the first draft, not me.”

Those generated quotes are not evidence of market prevalence, but they are good raw material for **affinity clustering, mental-model mapping, onboarding hypotheses, and trust-pattern design**.

What the synthetic sample suggests

The synthetic material points to **five recurring behavioral patterns** that are highly actionable for product direction.

First, users are rarely asking for “budgeting” in the abstract. They are asking for a **single coherent view** across fragmented money movement. This includes separate salary cards, online shopping cards, household cards, business cards, installment-related cards, and transfers between them. That pattern mirrors public requests in budgeting communities for tools that can track multiple cards, merchant histories, pending charges, and custom date ranges without forcing a perfect monthly-budget worldview. [7]

Second, the synthetic respondents repeatedly behaved like **“lightweight observers” rather than formal budgeters**. They do not want a full accounting system. They want timely awareness, low effort, and a gentle nudge before the damage is done. This is consistent with the appeal of tools like Revolut’s built-in analytics, PocketGuard’s overspend notifications, and Monzo’s trends and transaction context, all of which emphasize quick visibility over heavy setup. [8]

Third, the synthetic data strongly favored **payday-based and event-based views** over rigid calendar budgeting for freelancers, shift workers, parents, and small business owners. This tracks closely with real complaints that month boundaries distort real money management for biweekly earners, irregular earners, and people who budget around family events, annual expenses, or supplier cycles rather than clean monthly envelopes. [9]

Fourth, trust is less about jargon like “bank-grade security” and more about **concrete user control**. In the synthetic material, willingness to try notification-reading rose sharply when users were promised visible rules, easy corrections, local processing, manual fallback, data deletion, and permission reversibility. This aligns with Google Play’s consent requirements for personal and sensitive data access and with user discussions that specifically ask for manual or file-import alternatives. [10]

Fifth, users do not abandon trackers because they hate money management. They abandon them because the **maintenance burden arrives before the reward loop**. Manual entry, re-categorization, connector glitches, and waiting for delayed transaction sync all create work before the user sees insight. That same pattern appears repeatedly in the real-world corpus. [11]

Real-world signal mining

The real-world corpus I built for this project contains **112 atomic signal units** distilled from **52 public pages and threads** across Reddit, App Store pages, and Google Play pages. It spans YNAB, PocketGuard, Spendee, Revolut, Monzo, Goodbudget, Monarch-related discussions, and adjacent card-tracking/budgeting threads. To keep the dataset useful for affinity work and compliant with quotation limits, the downloadable sheet stores **short atomic user statements**, many of them **condensed or paraphrased** from longer comments while preserving the original sentiment and theme.

The pattern saturation across the public corpus is strong. Public users keep circling the same issues: manual entry helps awareness but does not scale; sync delays and duplicates erode trust; category engines are brittle; multi-account and transfer-heavy behavior breaks simplistic budget models; and privacy concerns push many people back toward spreadsheets, manual exports, or local-first tools. These patterns appear across app-store reviews and discussion forums, not just in one brand's feedback loop. [12]

A few especially revealing short quotes from the corpus:

"If I have to manually enter it, I'm thinking about it." [13]

"I just gave up on apps and use a Google sheet." [14]

"The Budget feature is a hot mess!" [15]

"I'm spending a lot more time on the app messing with categorizing transactions than I'd like to." [16]

These quotes matter because they reveal a subtle but important truth: users do not always want "more automation." They want **the right automation boundary**. YNAB discussions show that some users value manual entry because it increases awareness, while others want linked import only as a fallback. PocketGuard reviews praise estimated income and category alerts, but complain when transactions disappear. Spendee reviews praise multi-currency and visual clarity, but list missing rollover budgets, missing split transactions, and slow linked-account refresh. Revolut discussions show demand for editable analytics, exportability, and stable categories. Monarch threads repeatedly surface category regressions, duplicate transactions, annual-budget mismatch, and the need for custom rules and pay-cycle support. [17]

The downloaded signal sheet clusters into seven broad themes. The largest theme is **multi-account and model-fit needs**, followed by **sync reliability and bugs**, **privacy and trust**, **manual tracking and fallback tools**, and **categorization/data quality**. The smaller

but still important themes are **helpful automation/feedback** and **complexity/abandonment**.

Real-world signal units by theme

One important nuance from the corpus is that users often praise features in the same breath that they criticize the system. For example, Revolut's budgeting tools, PocketGuard's category alerts, Spendee's multi-currency support, and Monzo's trends/transaction metadata are all described as genuinely useful. But they lose impact when users hit stale refresh cycles, wrong categories, weak transfer handling, or uncertainty about what data third parties can see and retain. In other words, **the differentiator is not whether the product has analytics; it is whether users trust the numbers enough to act on them.** [18]

Product implications for an Iran-first notification-driven app

The strongest product direction emerging from both the synthetic and mined material is **automation with auditability**. Users clearly want the boring work removed, but they do not want opaque finance AI. The concept should therefore parse bank notifications automatically, show the extracted fields immediately, and make every category easy to correct with one tap. A visible rule like "future payments from this merchant = groceries" is more trustworthy than a hidden model decision. This direction is consistent with official app-positioning around auto-categorization and analytics, but also directly addresses the user backlash documented when categories drift, revert, or require endless custom cleanup. [19]

The second implication is **Android-first architecture with deliberate fallback modes**. On Android, notification access is the pragmatic route; SMS-reading is far more policy-constrained on Google Play. On iPhone, the same proposition should be repositioned as manual entry, file import, or bank-supported integrations rather than notification capture. For Iran specifically, where Android dominates mobile usage and banks vary in how they send transaction alerts, the product should support multiple capture modes: notification listener first, manual correction second, optional CSV or statement import third. That multi-path design is not feature bloat; it is how you preserve usefulness when banks, thresholds, or channels differ. [20]

The third implication is that the primary surface should be **spending awareness, not formal budgeting setup**. Users consistently respond well to quick-glance analytics, weekly summaries, overspend alerts, recurring-payment visibility, and easy merchant/category search. They respond poorly when the first experience asks for a full financial philosophy, a large category taxonomy, or complicated reconciliation concepts. The best early interaction model for this app is likely: permission → recent parsed transactions → editable categories → weekly digest → simple insights. Formal budgets can come later. [21]

The fourth implication is **pay-cycle flexibility and transfer literacy**. Users with irregular income, family budgeting, savings transfers, installment behavior, or multiple cards do not live inside neat monthly envelopes. The app therefore needs custom cycles such as payday-to-payday, explicit handling of internal transfers so they do not look like spending, and simple ways to separate household, personal, and business-like flows. If transfers and card payments are misread, the app will look “clever” but feel false. [22]

The fifth implication is **trust-by-design onboarding**. Because notification access is privacy-sensitive, the permission ask should explain exactly what is read, what is ignored, whether processing is on-device, how long data is kept, and how the user can revoke or delete everything. The experience should also explain why the app is requesting notification access instead of SMS access, and why some notifications may not be processed on all devices or banks. Given the security sensitivity around Iranian banking infrastructure in June 2026, the product should avoid vague reassurance and instead use concrete controls, transparent logs, and reversible permissions. [23]

A concise design translation of the research looks like this:

Recurrent signal	Product response
“I see transactions one by one, but not the pattern.”	Default to weekly summaries, category trends, and merchant rollups
“Manual logging keeps me aware, but I quit.”	Auto-capture first, minimal one-tap correction second
“Categories are wrong and I’m doing cleanup work.”	Show editable rules, confidence states, and quick recategorization
“My money is split across cards and transfers.”	Build multi-card merge logic and explicit transfer handling
“I don’t trust third parties with bank data.”	Use transparent permissioning, local-first positioning where possible, and manual/file-import fallback
“Real life is not aligned to calendar months.”	Support payday-based cycles, family views, and irregular-expense tracking

That table is a synthesis of the recurring friction and demand patterns documented in the public corpus and the synthetic sample. [24]

Research assets and limitations

This report is deliberately shaped to be **raw-material friendly** for downstream UX synthesis. The workbook can be used to build affinity diagrams, proto-personas, JTBD clusters, trust maps, onboarding hypotheses, and testable feature assumptions. The synthetic participant sheet is best used for **pattern exploration** and scenario writing. The real-world signal sheet is best used for **grounding those scenarios in public user language** and stress-testing which problems recur across tools.

There are also important limits. The synthetic dataset is simulated and therefore useful for discovery, not validation. The real-world corpus is composed of public comments and reviews, which over-represent people motivated enough to post, especially users with either strong positive or strong negative experiences. It is also skewed toward public English-language discussions rather than Persian-language app reviews or in-country Iranian banking communities. And because Iranian banks vary in how they deliver alerts, some of the most important product risks are not attitudinal at all but operational: message-format variability, bank-channel heterogeneity, Android permission friction, and trust concerns around notification access. [25]

Even with those limits, the signal is already strong enough to support an initial product thesis:

Users do not need another manual expense tracker. They need an Android-first, privacy-explicit, correction-friendly spending layer that turns fragmented bank notifications into one believable view of where the money went.

[1] [5] <https://corporateinsight.com/survey-how-and-why-consumers-budget/>

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[12] [13] [17] [24] https://www.reddit.com/r/ynab/comments/1fxwwyf/why_manually_log/

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